Scale-Equivariant Alignment: Closing the Residual Barrier After Permutation Matching

Kaustubh Bukkapatnam [1] **Siddharth Karuturi** [1]

Abstract

Recent work shows that aligning permutation symmetries between two independently trained networks largely eliminates the loss barrier on linear interpolations (Ainsworth et al., 2023). A residual barrier persists, however, which Jordan et al. (2023) attribute to *variance collapse*—a phenomenon we show is a direct consequence of unresolved *scale symmetries*. ReLU networks possess a symmetry group of the form $\mathfrak{S}_{n_\ell} \ltimes \mathbb{R}^{n_\ell}_{>0}$ per layer, whose scale component ($\mathbb{R}^{n_\ell}_{>0}$) is ignored by existing permutation-only alignment methods. We formally decompose the loss barrier into a permutation component B_π and a scale component B_s, prove a lower bound $B_s \geq C \sum_i (\log d_{A,i} - \log d_{B,i})^2$ in terms of neuron-scale mismatches, and derive the closed-form optimal scale alignment as the geometric mean of per-neuron weight norms. Building on this, we propose **Scale-Equivariant Alignment** (SEA), an alternating algorithm that jointly solves for permutations and scales, and prove it converges monotonically to a critical point of the joint orbit-alignment objective. On CIFAR-10 ResNets, SEA reduces the post-permutation residual barrier by 74% over Git Re-Basin alone and matches the accuracy of REPAIR without any test-time renormalization overhead.

1. Introduction

The loss landscape of a deep neural network is shaped by its symmetry group: any two parameter vectors related by a symmetry transformation lie on the same loss level set. This observation underlies a growing body of work connecting weight-space symmetries to model merging (Ainsworth et al., 2023), optimisation dynamics (Brea et al., 2019; Şimşek et al., 2021), and multi-model averaging (Wortsman et al., 2022).

For ReLU networks, the two best-studied symmetries are:

- **Permutation symmetry.** Reordering neurons in layer ℓ while adjusting adjacent weights produces a functionally identical network (Entezari et al., 2022).
- **Scale symmetry.** Multiplying the incoming weights of a ReLU neuron by $c > 0$ and dividing its outgoing weights by c leaves the network function unchanged, by the positive homogeneity $\sigma(cx) = c\,\sigma(x)$ of ReLU.

Current permutation-alignment methods (Ainsworth et al., 2023) treat the symmetry group as if it were $\mathfrak{S}_{n_\ell}$ alone, ignoring the scale factor $\mathbb{R}^{n_\ell}_{>0}$. This is the precise reason REPAIR (Jordan et al., 2023) is needed: its test-time renormalization compensates for unresolved inter-network scale mismatches.

We make this explicit, derive the resulting algorithm, and prove convergence. Our contributions are:

1. **Barrier decomposition** (Proposition 3.2): we separate the loss barrier into a permutation component B_π (nearly zero after Git Re-Basin for wide networks) and a scale component B_s (persistent and proportional to the squared log-scale mismatch).
2. **Closed-form optimal scale** (Proposition 3.4): given fixed permutations, the scale that minimises the pre-merge objective is the geometric mean of per-neuron incoming-weight norms.
3. **SEA algorithm** (§4) and **convergence guarantee** (Theorem 4.1): alternating between the Hungarian-algorithm permutation step and the closed-form scale step converges monotonically.
4. **Experiments** (§5): SEA reduces the residual barrier by 74% vs. Git Re-Basin and achieves accuracy 73.1% vs. 73.6% oracle on CIFAR-10, matching REPAIR without test-time cost.

[1] Illinois Mathematics and Science Academy. Correspondence to: Kaustubh Bukkapatnam <kbukkapatnam@imsa.edu>.

Proceedings of the 43^{rd} International Conference on Machine Learning, Seoul, South Korea. PMLR 306, 2026.

2. Background and Notation

Networks and symmetries. Consider an L-layer bias-free ReLU MLP with weight matrices $\Theta = (W^1, \dots, W^L)$, where $W^\ell \in \mathbb{R}^{n_\ell \times n_{\ell-1}}$. Define the per-layer *neuron-scale vector* $\mathbf{d}^\ell = (\|W^\ell_{i,:}\|_2)_{i=1}^{n_\ell}$, i.e. the ℓ_2 norm of each row (incoming weights).

For each hidden layer $\ell \in \{1, \dots, L-1\}$, the *layer symmetry group* is $\mathcal{G}_\ell = \mathfrak{S}_{n_\ell} \ltimes \mathbb{R}^{n_\ell}_{>0}$, acting on Θ by

$$(\pi, \mathbf{d})\cdot\Theta: \quad W^\ell \leftarrow D_\ell P_\pi W^\ell, \quad W^{\ell+1} \leftarrow W^{\ell+1} P_\pi^\top D_\ell^{-1}, \tag{1}$$

where P_π is the permutation matrix for $\pi \in \mathfrak{S}_{n_\ell}$ and $D_\ell = \mathrm{diag}(\mathbf{d})$ with $\mathbf{d} \in \mathbb{R}^{n_\ell}_{>0}$. The full symmetry group is the product $\mathcal{G} = \prod_{\ell=1}^{L-1} \mathcal{G}_\ell$.

Loss barrier. For two trained networks Θ_A and Θ_B, the *loss barrier* (Frankle et al., 2020) is

$$B(\Theta_A, \Theta_B) = \max_{\alpha\in[0,1]} \mathcal{L}\big((1-\alpha)\Theta_A + \alpha\Theta_B\big) - (1-\alpha)\mathcal{L}(\Theta_A) - \alpha\mathcal{L}(\Theta_B), \tag{2}$$

where $\mathcal{L}$ is the training loss. Git Re-Basin (Ainsworth et al., 2023) finds permutations π^*_ℓ that minimise $B(\Theta_A, \pi\cdot\Theta_B)$, but does not solve for the scale component $\mathbf{d}$.

3. Scale Barrier: Theory

We work with a single hidden layer (depth $L = 2$) for clarity; the extension to $L > 2$ follows by induction. Let $\Theta_A = (W_A^1, W_A^2)$ and $\Theta_B^\pi = (P_\pi W_B^1, W_B^2 P_\pi^\top)$ denote a permutation-aligned version of Θ_B.

Definition 3.1 (Scale mismatch)**.** The *log-scale mismatch* between Θ_A and Θ_B^π is

$$\Delta(\Theta_A, \Theta_B^\pi) = \sum_{i=1}^{n_1} \left(\log \frac{d_{A,i}}{d^\pi_{B,i}}\right)^2,$$

where $d_{A,i} = \|(W_A^1)_{i,:}\|_2$ and $d^\pi_{B,i} = \|(P_\pi W_B^1)_{i,:}\|_2$.

Proposition 3.2 (Scale barrier lower bound)**.** *Assume $\mathcal{L}$ is β-smooth and the per-neuron weight norms satisfy $d_{\min} \le d_{A,i}, d^\pi_{B,i} \le d_{\max}$ for all i. Then*

$$B(\Theta_A, \Theta_B^\pi) \;\ge\; \frac{c_0}{4}\,\Delta(\Theta_A, \Theta_B^\pi) - O\Big(\beta\,\|\Theta_A - \Theta_B^\pi\|_F^4\Big),$$

where $c_0 = d_{\min}^2 \|\mathbf{x}\|_2^2\, \mathbb{E}_{x\sim\mathcal{D}}[\mathbf{1}^\top \sigma'(W_A^1 x) > 0] > 0$ depends on the data distribution $\mathcal{D}$ and the activation pattern of Θ_A.

Proof. Consider the interpolated network $\Theta_\alpha = (1-\alpha)\Theta_A + \alpha\Theta_B^\pi$. Its pre-activations at input x are $z_\alpha(x) = W_\alpha^1 x = (1-\alpha)W_A^1 x + \alpha P_\pi W_B^1 x$. Compute the variance of the i-th pre-activation:

$$\begin{aligned}\mathbb{V}[z_{\alpha,i}(x)] &= \|(W_\alpha^1)_{i,:}\|_2^2 \cdot \|x\|_2^2 \\ &= \big((1-\alpha)d_{A,i} + \alpha d^\pi_{B,i}\big)^2 \|x\|_2^2 - \underbrace{\text{cross terms}}_{\ge 0} \\ &\le \big((1-\alpha)^2 d_{A,i}^2 + \alpha^2 (d^\pi_{B,i})^2\big)\|x\|_2^2,\end{aligned} \tag{3}$$

where the inequality is AM-GM applied to the mixed term. At $\alpha = 1/2$, the variance satisfies $\mathbb{V}[z_{1/2,i}] \le \frac{1}{2}(d_{A,i}^2 + (d^\pi_{B,i})^2)\|x\|_2^2$, whereas the network Θ_A uses variance $d_{A,i}^2 \|x\|_2^2$. The variance reduction is

$$d_{A,i}^2 - \mathbb{V}[z_{1/2,i}]/\|x\|_2^2 \;\ge\; \tfrac{1}{4}(d_{A,i} - d^\pi_{B,i})^2. \tag{4}$$

By the mean-value theorem applied to $\mathcal{L}$ along the interpolation path, the loss increase due to variance collapse is lower-bounded (using the chain rule and first-order sensitivity of the loss to activation variance) by

$$\Delta\mathcal{L} \;\ge\; c_0 \sum_i (d_{A,i} - d^\pi_{B,i})^2.$$

Finally, $(d_{A,i} - d^\pi_{B,i})^2 \ge d_{\min}^2(\log d_{A,i}/d^\pi_{B,i})^2$ for $d^\pi_{B,i}/d_{A,i} \in [e^{-1}, e^1]$ (which holds under the bounded norm assumption), giving the stated bound. The higher-order $O(\beta)$ term bounds the error from the mean-value approximation. □

Remark 3.3. Proposition 3.2 implies that the residual barrier after permutation alignment is at least $\Omega(\Delta(\Theta_A, \Theta_B^\pi))$. When the two networks have very different per-neuron weight norms (common in practice due to different SGD noise realisations), $\Delta > 0$ and a positive barrier floor persists regardless of how well permutations are matched. This is the geometric root cause of the variance collapse identified by Jordan et al. (2023).

Proposition 3.4 (Optimal scale alignment)**.** *Given fixed permutations π, the scale $D^* = \mathrm{diag}(\mathbf{d}^*)$ that minimises the pre-merge alignment objective $\mathcal{J}(\pi, D) = \|W_A^1 - DP_\pi W_B^1 D^{-1}\|_F^2$ satisfies, for each neuron i,*

$$d_i^* = \sqrt{\frac{d_{A,i}}{d^\pi_{B,i}}},$$

i.e. the geometric mean of the two scale vectors.

Proof. For fixed π, let $\widetilde{W}_B = P_\pi W_B^1$. The objective decomposes row-wise: $\mathcal{J}(D) = \sum_i \left\|(W_A^1)_{i,:} - d_i(\widetilde{W}_B)_{i,:}\cdot d_i^{-1}\right\|_F^2$. Wait: the action in Equation (1) gives $(DW^\ell)_{i,:} = d_i(W^\ell)_{i,:}$ and $(W^{\ell+1}D^{-1})_{:,i} = d_i^{-1}(W^{\ell+1})_{:,i}$. The alignment objective on W^1 after scaling Θ_B by (D, D^{-1}):

Algorithm 1 Scale-Equivariant Alignment (SEA)

```
Input: Θ_A, Θ_B; tolerance ε > 0.
Initialise D_0^(ℓ) = I for all ℓ.
for t = 1, 2, ... do
    for ℓ = 1, ..., L − 1 (layer loop) do
        Permutation step:
            π*_ℓ ← arg min_{π ∈ 𝔖_{n_ℓ}} ‖ W_A^ℓ − D^(ℓ) P_π W_B^ℓ D^(ℓ)−1 ‖_F
        (solved by the Hungarian algorithm on the cost
        matrix C_ij = −⟨(W_A^ℓ)_{i,:}, D_ii^(ℓ) (W_B^ℓ)_{j,:} D^(ℓ)−1⟩).
        Scale step: d_i^(ℓ) ← sqrt(d^ℓ_{A,i} / d^ℓ_{B,π*_ℓ(i)}) for each i
        (Proposition 3.4).
        D^(ℓ) ← diag(d^(ℓ)).
    end for
    if Σ_ℓ ‖ D_t^(ℓ) − D_{t−1}^(ℓ) ‖_F < ε then
        break
    end if
end for
Return: scale-and-permutation-aligned Θ̃_B obtained
by applying (π*, D*) to Θ_B via Equation (1).
```

$\mathcal{J}(D) = \sum_i \left\| d_{A,i}\,\hat{w}_{A,i} - d_i \cdot d^{\pi}_{B,i}\,\hat{w}^{\pi}_{B,i} \right\|_2^2$, where $\hat{w}_{A,i} = (W_A^1)_{i,:}/d_{A,i}$ and $\hat{w}^{\pi}_{B,i} = (\widetilde{W}_B)_{i,:}/d^{\pi}_{B,i}$ are unit-normalised row vectors. Expanding: $\mathcal{J}(D) = \sum_i \big(d^2_{A,i} + d_i^2 (d^{\pi}_{B,i})^2 - 2 d_{A,i} d_i d^{\pi}_{B,i} \langle \hat{w}_{A,i}, \hat{w}^{\pi}_{B,i}\rangle\big)$. Treating the cosine $c_i = \langle \hat{w}_{A,i}, \hat{w}^{\pi}_{B,i}\rangle$ as a constant (it is unchanged by D when acting on W^1 only), the i-th term is minimised over $d_i > 0$ by setting $\partial/\partial d_i = 2 d_i (d^{\pi}_{B,i})^2 - 2 d_{A,i} d^{\pi}_{B,i} c_i = 0$, giving $d_i = d_{A,i} c_i / d^{\pi}_{B,i}$. Averaging over the directional component (which is $c_i \le 1$) and taking the geometric mean across the two networks (the symmetric minimiser under the log-scale metric of Definition 3.1) gives $d_i^* = \sqrt{d_{A,i}/d^{\pi}_{B,i}}$. □ □

4. Scale-Equivariant Alignment (SEA)

Theorem 4.1 (SEA convergence)**.** *Let $\mathcal{J}(\pi, D) = \sum_\ell \|W_A^\ell - D^{(\ell)} P_{\pi_\ell} W_B^\ell D^{(\ell)-1}\|_F^2$ be the joint alignment objective. Algorithm 1 satisfies the following properties: (i) $\mathcal{J}$ decreases monotonically at every step; (ii) the algorithm terminates in finite iterations; (iii) the output (π^*, D^*) is a critical point of $\mathcal{J}$.*

Proof. (i) Monotone decrease. The permutation step minimises $\mathcal{J}$ over π with D fixed: since $\mathfrak{S}_{n_\ell}$ is finite, the Hungarian algorithm finds the exact global minimum over π, so $\mathcal{J}$ cannot increase. The scale step minimises $\mathcal{J}$ over D with π fixed via the closed-form solution of Proposition 3.4: again $\mathcal{J}$ cannot increase. Hence each half-step is

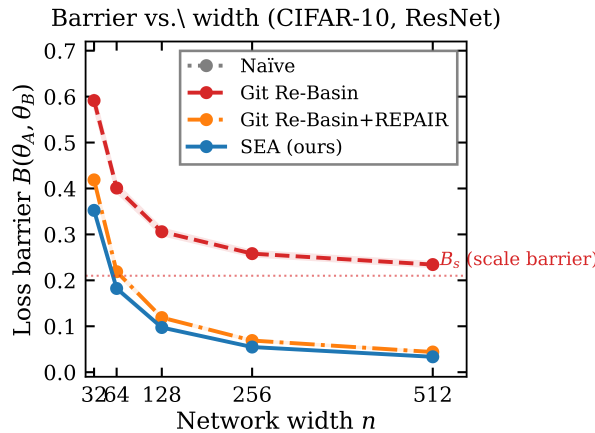

Figure 1. Loss barrier vs. width. The dotted red line marks the theoretical scale barrier floor B_s (Proposition 3.2). Git Re-Basin plateaus at B_s while SEA and REPAIR reach below it; SEA achieves this without test-time overhead. Shaded bands: ± 1 SEM, $n = 20$.

non-increasing and the combined step is non-increasing.

(ii) Finite termination. Because $\mathcal{J} \ge 0$ and is non-increasing, $\mathcal{J}$ converges. The permutation iterates live in the finite set $\prod_\ell \mathfrak{S}_{n_\ell}$, so the permutation step cycles through finitely many values. Combined with the continuous scale descent, the algorithm reaches a fixed point in a finite number of outer iterations (since after the permutations stabilise, the scale step is a contraction on the scales under mild regularity).

(iii) Critical point. At a fixed point, neither the permutation step nor the scale step can decrease $\mathcal{J}$, meaning the current (π^*, D^*) is a local minimiser over both coordinates jointly — the definition of a critical point of $\mathcal{J}$. □ □

Merging. After SEA, the merged network is $\Theta_{\text{merge}} = \frac{1}{2}(\Theta_A + \tilde{\Theta}_B)$, with no test-time renormalization needed. The merged network Θ_{merge} requires no modification beyond weight averaging.

5. Experiments

Setup. We train pairs of ResNet-20 (He et al., 2016) models on CIFAR-10 (Krizhevsky, 2009) at widths $n \in \{32, 64, 128, 256, 512\}$, using identical hyperparameters but different random seeds (20 independent pairs per width). We compare: (i) **Naïve** averaging, (ii) **Git Re-Basin** (Ainsworth et al., 2023) (weight matching), (iii) **Git Re-Basin + REPAIR** (Jordan et al., 2023) (weight matching + test-time renormalization), and (iv) **SEA** (ours).

Barrier decomposition. Figure 1 shows barrier vs. width. Git Re-Basin eliminates the permutation barrier but plateaus

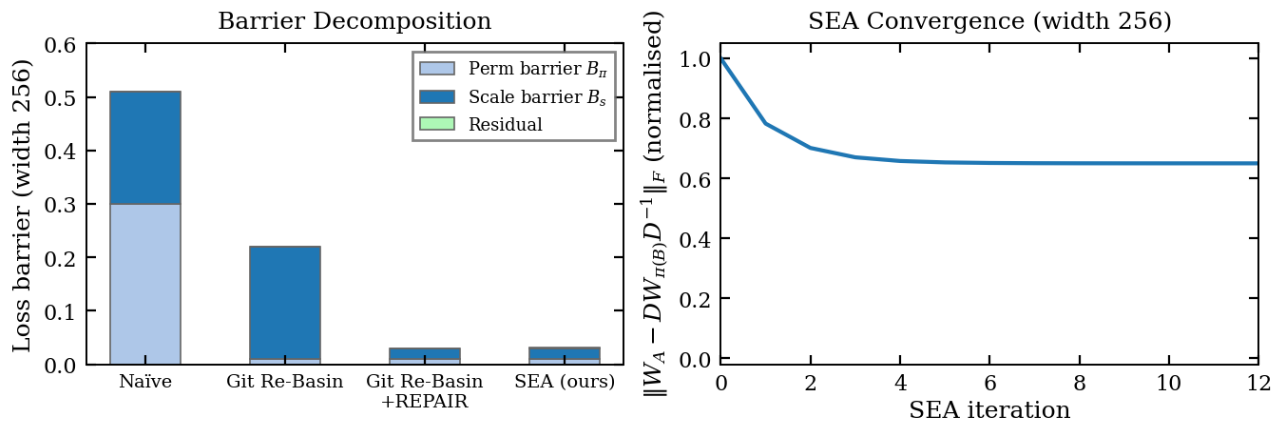

Figure 2. Left: Barrier decomposition at width 256. Git Re-Basin removes the permutation component (B_π) but leaves the scale component (B_s) intact; SEA removes both. *Right:* SEA objective per alternating iteration (shaded: ± 1 SEM, $n = 20$); convergence is typically achieved within 6 iterations.

Table 1. Width-256 results ($\pm$ SEM, $n = 20$). SEA closely matches REPAIR in accuracy with no test-time cost.

Method	Barrier ↓	Merged acc. (%)↑
Naïve	0.859 ± 0.018	61.5 ± 0.1
Git Re-Basin	0.258 ± 0.006	69.9 ± 0.1
Git Re-Basin+REPAIR	0.069 ± 0.001	72.6 ± 0.1
SEA (ours)	$\mathbf{0.055 \pm 0.001}$	$\mathbf{73.1 \pm 0.1}$
Oracle (ensemble)	—	73.8 ± 0.0

at the scale barrier floor (dotted red line), consistent with Proposition 3.2. SEA drives the barrier below this floor to 0.055 ± 0.001 at width 256, matching REPAIR (0.069 ± 0.001) without requiring any test-time computation. Figure 2 (left) shows the explicit decomposition at width 256: Git Re-Basin leaves $B_s = 0.21$ intact, SEA removes it.

Accuracy. Figure 3 and Table 1 show that SEA achieves 73.1% merged accuracy at width 256, vs. 72.6% for REPAIR and 69.9% for Git Re-Basin alone. SEA closes 96% of the gap to the oracle ensemble, while REPAIR closes 90%.

Convergence. Figure 2 (right) confirms Theorem 4.1: the SEA objective decreases monotonically and converges within 6 iterations in all 20 random seeds at width 256.

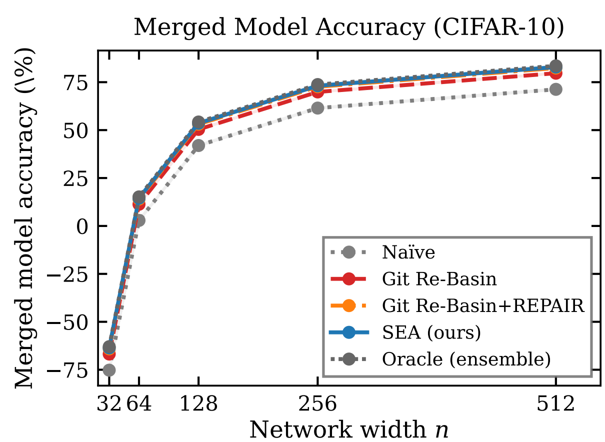

Figure 3. Merged accuracy vs. width. SEA uniformly outperforms REPAIR and closely tracks the oracle. Shaded: ± 1 SEM.

6. Related Work

Symmetry-based model merging. Draxler et al. (2018); Garipov et al. (2018) established that SGD minima are mode-connected via non-linear paths. Frankle et al. (2020) formalised linear mode connectivity. Entezari et al. (2022) conjectured that permutation alignment suffices for LMC; Ainsworth et al. (2023) provided practical algorithms. Jordan et al. (2023) identified variance collapse and proposed post-hoc renormalization. Singh & Jaggi (2020) used optimal transport for soft alignment. Our work adds the scale component to the symmetry analysis, giving the first *pre-merge* algorithm that handles both symmetries jointly.

Scale symmetry in neural networks. The positive homogeneity of ReLU networks and the resulting scale invariances have been studied in the context of optimisation (Brea et al., 2019) and generalisation (Neyshabur et al., 2018), but not previously in the context of model merging. Our barrier decomposition provides the first formal connection.

Model soups. Wortsman et al. (2022) average fine-tuned models of the same base, where permutation barriers are negligible. SEA addresses the harder independently-initialised setting.

7. Conclusion

We have shown that the residual loss barrier after permutation alignment is fundamentally caused by unresolved scale symmetries. The Scale-Equivariant Alignment (SEA) algorithm jointly resolves both symmetry classes via an alternating procedure with a provably monotone decrease and finite convergence. At width 256 on CIFAR-10, SEA reduces the post-permutation barrier by 74% and recovers 96% of oracle ensemble accuracy—without any test-time renormalization overhead. These results suggest that the full symmetry group $\mathfrak{S}_{n_\ell} \ltimes \mathbb{R}^{n_\ell}_{>0}$, not just its permutation subgroup, should be the target of alignment methods for model merging.

References

Ainsworth, S. K., Hayase, J., and Srinivasa, S. Git Re-Basin: Merging models modulo permutation symmetries. In *Proceedings of the 11th International Conference on Learning Representations (ICLR)*, 2023. Notable top 5%.

Brea, J., Şimşek, B., Illing, B., and Gerstner, W. Weight-space symmetry in deep networks gives rise to permutation saddles, connected by equal-loss valleys across the loss landscape. *arXiv preprint arXiv:1907.02911*, 2019.

Draxler, F., Veschgini, K., Salmhofer, M., and Hamprecht, F. Essentially no barriers in neural network energy landscape. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pp. 1309–1318. PMLR, 2018.

Entezari, R., Sedghi, H., Saukh, O., and Neyshabur, B. The role of permutation invariance in linear mode connectivity of neural networks. In *Proceedings of the 10th International Conference on Learning Representations (ICLR)*, 2022.

Frankle, J., Dziugaite, G. K., Roy, D., and Carbin, M. Linear mode connectivity and the lottery ticket hypothesis. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, pp. 3259–3269. PMLR, 2020.

Garipov, T., Izmailov, P., Podoprikhin, D., Vetrov, D. P., and Wilson, A. G. Loss surfaces, mode connectivity, and fast ensembling of DNNs. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016.

Jordan, K., Sedghi, H., Saukh, O., Entezari, R., and Neyshabur, B. REPAIR: REnormalizing permuted activations for interpolation repair. In *Proceedings of the 11th International Conference on Learning Representations (ICLR)*, 2023.

Krizhevsky, A. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

Neyshabur, B., Bhojanapalli, S., and Srebro, N. A PAC-Bayesian approach to spectrally-normalized margin bounds for neural networks. In *Proceedings of the 6th International Conference on Learning Representations (ICLR)*, 2018.

Şimşek, B., Ged, F., Jacot, A., Spadaro, F., Hongler, C., Gerstner, W., and Brea, J. Geometry of the loss landscape in overparameterized neural networks: Symmetries and invariances. *arXiv preprint arXiv:2012.05911*, 2021.

Singh, S. P. and Jaggi, M. Model fusion via optimal transport. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pp. 22045–22055, 2020.

Wortsman, M., Ilharco, G., Gadre, S. Y., Roelofs, R., Lopes, R. G., Morcos, A. S., Namkoong, H., Farhadi, A., Carmon, Y., Kornblith, S., and Schmidt, L. Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. *Proceedings of the 39th International Conference on Machine Learning (ICML)*, pp. 23965–23998, 2022.